

Solution first chapter Mathematical methods for physicists

Abstract

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1 Exercise 1.1

1. Prove that if $\lim_{n \rightarrow \infty} n^p u_n = A < \infty$ with $p > 1$, then the series $\sum_{n=1}^{\infty} u_n$ converges.
 2. Prove that if $\lim_{n \rightarrow \infty} n u_n = A > 0$, then the series diverge (the test fails for $A = 0$).
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2 Exercise 1.2

If $\lim_{n \rightarrow \infty} \frac{b_n}{a_n} = K$ a constant $0 < K < \infty$, show that $\sum_n b_n$ converges or diverges with a_n .
